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Course Code: BTD50112
Course Name: Computer Networks

PBL Assignment – 2

Title: Reliable data transfer system for online banking

Objective

Analyse & implement suitable error detection, correction, flow control & error control mechanisms for reliable transmission of banking transaction data.

Real Life Connection

An online banking system must transfer financial transaction data securely. A corrupted frame could result in incorrect transaction information; therefore, the network requires appropriate error detection, correction and transmission mechanisms to make a secure connection between client and server.

Target

- (i) Design the network for secure banking transaction among client and server using Cisco Packet Tracer.
- (ii) Consider the data packet binary sequence is 1101011011; the generator polynomial is 10011. Identify the CRC code and demonstrate how the receiver detects an error. Recalculate the CRC at the receiver.
- (iii) Determine whether the frame is accepted or rejected by the end user according to design.

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1. Objective

The objective is to design and describe a Reliable Data Transfer System for Online Banking that supports accurate transmission of banking transaction data between clients and a bank server. The project combines a routed Cisco Packet Tracer network with CRC-based error detection and a clear receiver-side accept/reject decision.

- Design a client-server banking network using Cisco Packet Tracer.
- Implement CRC (Cyclic Redundancy Check) for detecting transmission errors.
Demonstrate how a receiver identifies a corrupted frame.
- Recalculate and verify the CRC at the receiver.
- Determine whether a frame is accepted or rejected according to the CRC result.
- Explain how retransmission/error-control mechanisms can be used to recover from detected errors.

2. Real-Life Connection and Project Targets

Online banking transactions carry information that must arrive accurately. A corrupted bit in a transaction frame can change the meaning of the transmitted data. Error detection therefore provides an important reliability layer. CRC is used in this project to detect transmission errors; it does not itself correct a corrupted bit.

Project Targets

Target	Implementation in this project
(i) Network design	Client LAN → Router 1 → serial link → Router 2 → Bank Server LAN in Cisco Packet Tracer.
(ii) CRC	For data 1101011011 and generator 10011, calculate the CRC and show receiver verification.
(iii) Decision	Use the receiver's CRC remainder: 0000 → accept; non-zero → reject.

3. Introduction

The network model contains four client PCs connected to a switch on the client side and a bank server connected to a second switch on the server side. Two routers provide Layer-3 communication between the LANs, with a /30 serial network between the routers. The Packet Tracer model represents the communication path; CRC is treated as the logical error-detection mechanism for a transmitted data frame.

The design is a simplified academic model. It demonstrates reliable transfer concepts rather than the complete security architecture of a production banking system. Encryption, authentication, application-level banking services, and automated retransmission are identified as enhancements rather than claimed as implemented features.

4. System Model and Network Design

4.1 Network Topology

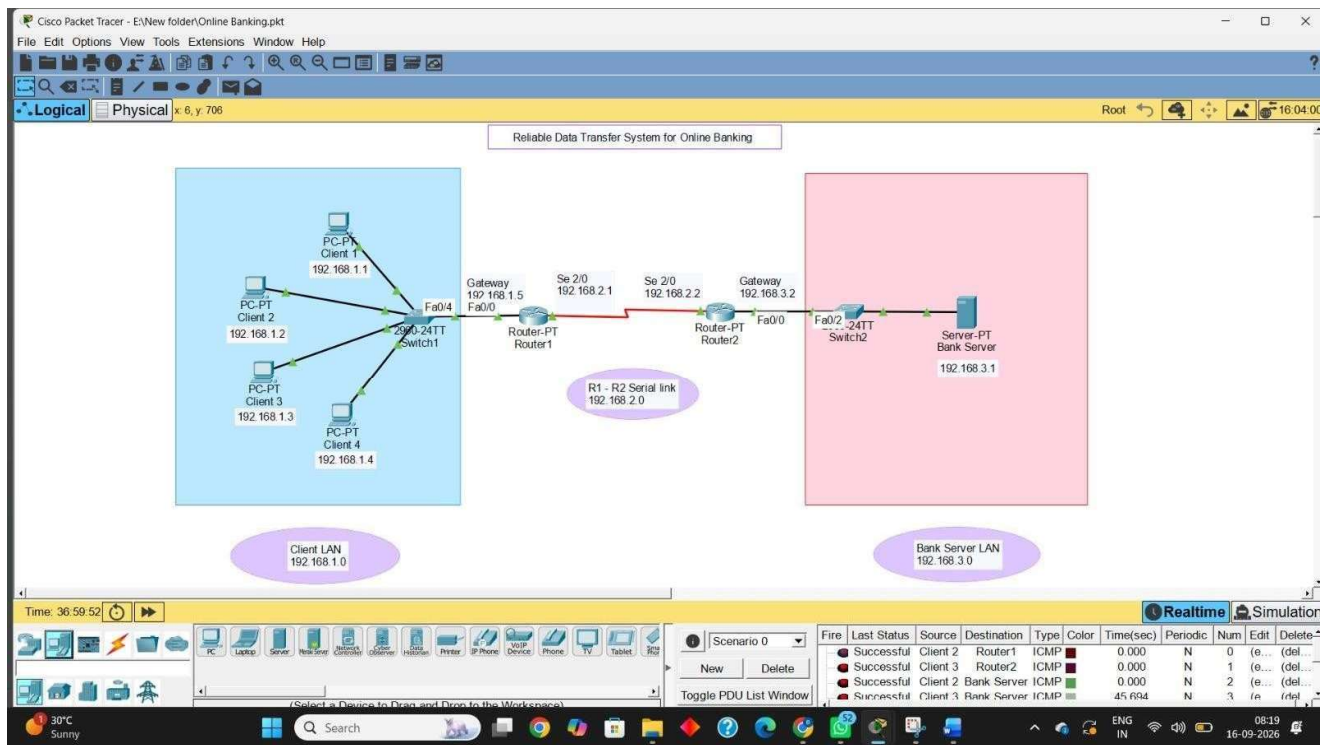


Figure 1. Cisco Packet Tracer topology used for the project.

4.2 IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0 / Client 1	FastEthernet0	192.168.1.1	255.255.255.0	192.168.1.5
PC1 / Client 2	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.5
PC2 / Client 3	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.5
PC3 / Client 4	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.5
Router1	FastEthernet0/0	192.168.1.5	255.255.255.0	—
Router1	Serial2/0	192.168.2.1	255.255.255.252	—
Router2	Serial2/0	192.168.2.2	255.255.255.252	—
Router2	FastEthernet0/0	192.168.3.2	255.255.255.0	—
Server0 / Bank Server	FastEthernet0	192.168.3.1	255.255.255.0	192.168.3.2

4.3 The Three Networks

Network	Network Address	Purpose
Client LAN	192.168.1.0/24	Connects PC0–PC3 to Router1.
R1–R2 Serial Link	192.168.2.0/30	Point-to-point router connection.
Bank Server LAN	192.168.3.0/24	Connects Router2 to the bank server.

The /30 serial network provides two usable host addresses: 192.168.2.1 for Router1 and 192.168.2.2 for Router2.

4.4 Packet Flow

Client PC → Switch0 → Router1 Fa0/0 → Router1 Serial2/0 → Router2 Serial2/0 → Router2 Fa0/0 → Switch1 → Bank Server

The client-side default gateway is 192.168.1.5. The bank-server-side default gateway is 192.168.3.2.

5. Cisco Packet Tracer Configuration

5.1 PC and Server Configuration

Each client is configured in the 192.168.1.0/24 network with Router1 (192.168.1.5) as its default gateway. The bank server uses 192.168.3.1/24 with Router2 (192.168.3.2) as its default gateway.

5.2 Router Interfaces and Routing

The router interfaces follow the addressing table above. For the two LANs to communicate, the routers require routes to the remote LANs. A static-routing configuration can be used in this topology:

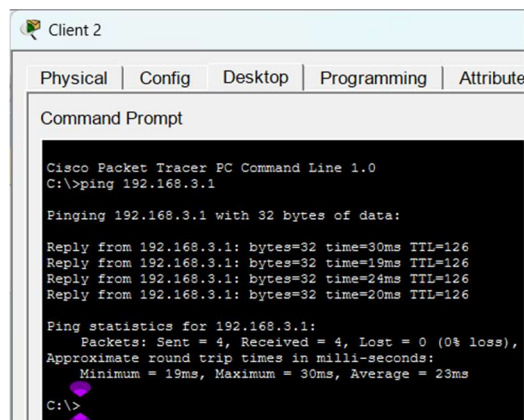
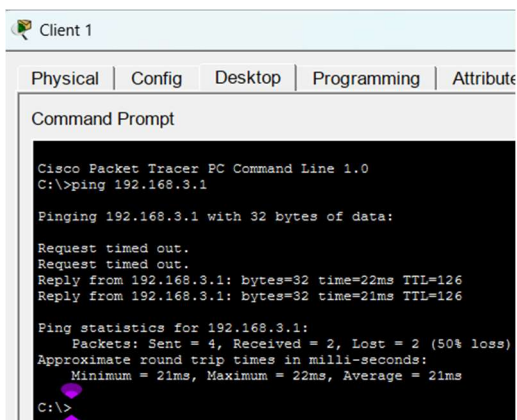
```
Router1:
ip route 192.168.3.0 255.255.255.0 192.168.2.2

Router2:
ip route 192.168.1.0 255.255.255.0 192.168.2.1
```

Because the serial connection is DCE on one side, the DCE interface also requires a clock rate in Packet Tracer. The exact clock-rate value is not stated in the supplied report, so it should be verified from the .pkt file before documenting a specific value.

5.3 Connectivity Testing

The supplied project materials report successful ICMP communication through the topology. A suitable end-to-end verification is:



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=31ms TTL=126
Reply from 192.168.3.1: bytes=32 time=41ms TTL=126
Reply from 192.168.3.1: bytes=32 time=24ms TTL=126
Reply from 192.168.3.1: bytes=32 time=18ms TTL=126

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 41ms, Average = 28ms
C:\>

```

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=31ms TTL=126
Reply from 192.168.3.1: bytes=32 time=17ms TTL=126
Reply from 192.168.3.1: bytes=32 time=19ms TTL=126
Reply from 192.168.3.1: bytes=32 time=15ms TTL=126

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 31ms, Average = 20ms
C:\>

```

6. CRC-Based Error Detection

6.1 Given Data and Generator

Parameter	Value
Data sequence	1101011011
Generator	10011
Generator length	5 bits
Generator degree	4
Zeros appended	4

Since the generator has length 5, four zeros are appended to the data before modulo-2 division

```

Data           = 1101011011
Padded data    = 11010110110000
Generator polynomial = 10011

```

6.2 CRC Calculation

CRC uses modulo-2 (XOR) division. The padded data 11010110110000 is divided by 10011. The final four bits of the division remainder form the CRC.

Modulo-2 division:

```

Dividend = 11010110110000
Generator = 10011

```

After XOR division, the 4-bit remainder is:

```
CRC = 1110
```

Therefore, the CRC code is:

CRC = 1110

A compact verification can also be expressed as:

$11010110110000 \div 10011 \rightarrow \text{remainder } 1110$

6.3 Transmitted Frame

The sender appends the CRC to the original data:

```
Original data = 1101011011
CRC          = 1110
-----
Transmitted  = 11010110111110
```

Transmitted frame = 11010110111110

7. Receiver-Side Verification

7.1 Correct Frame

At the receiver, the complete received codeword is divided by the same generator 10011.

```
Received frame = 11010110111110
Generator      = 10011
Remainder      = 0000
```

A zero remainder means no error is detected by this CRC check.

0000 → FRAME ACCEPTED

7.2 Corrupted Frame

To demonstrate error detection, assume one bit is changed during transmission. For example, the first bit of the transmitted frame changes from 1 to 0:

```
Original frame = 11010110111110
Corrupted frame = 01010110111110
```

Dividing the corrupted frame by the same generator gives:

```
Received frame = 01010110111110
Generator      = 10011
Remainder      = 0100
```

Because 0100 is non-zero, the receiver detects an error.

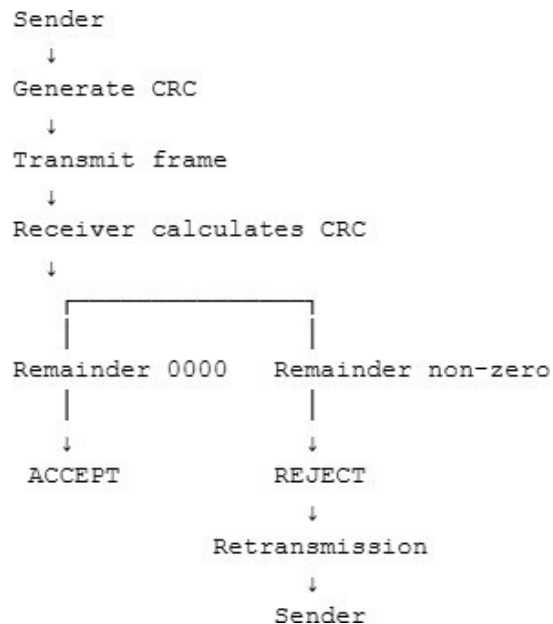
0100 ≠ 0000 → ERROR DETECTED → FRAME REJECTED

7.3 Acceptance/Rejection Rule

Receiver CRC remainder	Interpretation	End-user decision
0000	No transmission error detected by CRC	ACCEPT
Non-zero (e.g., 0100)	Transmission error detected	REJECT / request retransmission

CRC detects errors; it does not correct the corrupted bit. A reliable system can use an error-control protocol such as ARQ to request retransmission after rejection.

8. Error Control and Reliable Transfer



This separates the roles of the mechanisms: CRC provides error detection, while an ARQ-style retransmission mechanism provides recovery from detected corruption. Flow control and full application-level banking logic are outside the demonstrated Packet Tracer implementation.

9. Results and Observation

Test	Input/Condition	CRC result	Decision
Normal transmission	11010110111110	0000	Accepted
Single-bit corruption example	01010110111110	0100	Rejected

- The client LAN is 192.168.1.0/24 and the bank server LAN is 192.168.3.0/24. Router1 and Router2 are connected through the 192.168.2.0/30 serial network.
- The supplied Packet Tracer results show successful end-to-end ICMP communication. The calculated CRC for data 1101011011 with generator 10011 is 1110.
- The receiver obtains remainder 0000 for the correct frame and a non-zero remainder for the demonstrated corrupted frame.

10. Limitations

- The current Packet Tracer model demonstrates network connectivity and the project's CRC logic; it is not a full production banking system.
- CRC is an error-detection mechanism and does not by itself correct corrupted data. Encryption and user authentication are not implemented in the supplied topology.

- ARQ/retransmission is described as the recovery mechanism rather than demonstrated as an application protocol in Packet Tracer.
- ICMP ping verifies IP connectivity but does not simulate a complete banking transaction such as fund transfer or balance update.

11. Conclusion

The Reliable Data Transfer System for Online Banking has been designed using Cisco Packet Tracer with a client LAN, two routers, a serial inter-router link, and a bank-server LAN. The network addressing and routed path provide end-to-end connectivity between the clients and bank server. The CRC portion of the project demonstrates how the data sequence 1101011011 is encoded using generator 10011, producing CRC 1110 and the transmitted frame 1101011011110. At the receiver, a remainder of 0000 results in acceptance, while the demonstrated corrupted frame produces a non-zero remainder (0100) and is rejected. The project therefore demonstrates the requested error-detection and acceptance/rejection logic, while ARQ, encryption, authentication, and a full banking application remain natural extensions.

12. Future Scope

- Implement ARQ-based automatic retransmission of rejected frames.
- Add checksum, Hamming code, or stronger CRC variants for comparative study. Add encryption and authentication for confidentiality and identity verification.
- Build a real-time client-server banking application for balance inquiry, fund transfer, and payment processing. Measure packet loss, delay, bandwidth, and error rates under different network conditions.
- Scale the topology to multiple branches, servers, and redundant links. Explore cloud-based deployment and high-availability designs.

13. References

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